

Bhumika Dommaraju

bhumikadommaraju7@gmail.com | +1 (518) 870-7737 | New York, USA | [LinkedIn](#)

Profile Summary

Software Engineer with 4 years of experience building enterprise-grade financial and healthcare platforms using Java, Spring Boot, React, Docker, Kubernetes, and cloud-native architectures. Skilled in designing scalable microservices, optimizing high-throughput APIs, and engineering secure, compliant workflows aligned with regulatory standards. Adept in distributed systems, event-driven messaging, SQL/NoSQL optimization, and modern frontend engineering. Proven ability to enhance system performance, strengthen platform reliability, and deliver data-driven features powering transactional, reporting, and clinical integration systems.

Technical Skills

Programming Languages & Core Technologies: Java, JavaScript (ES6+), TypeScript, Python, C, C++, SQL, Shell Scripting, HTML5, CSS3, JSON, XML

Backend Engineering & Frameworks: Spring Boot, Spring MVC, Spring Security, Hibernate/JPA, Microservices Architecture, RESTful APIs, GraphQL, Node.js, Express.js, gRPC, Event-Driven Services, JWT/OAuth2

Frontend Engineering & UI Ecosystem: React.js, angular, Next.js, Redux Toolkit, Material UI, Tailwind CSS, Webpack, Babel, Responsive UI Engineering, Component-Driven Architecture, Single-Page Applications (SPA)

Cloud, DevOps & Platform Engineering: AWS (ECS, Lambda, S3, IAM, DynamoDB, RDS), Azure App Services, Google Cloud Run, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD Orchestration, API Gateways, Serverless Workflows

Data, Messaging & Storage Systems: PostgreSQL, MySQL, MongoDB, Redis, Kafka, RabbitMQ, Elasticsearch, Cassandra, API Integration, Distributed Caching, Stream Processing, ORM Optimization

Software Engineering Tools & Practices: Git, GitLab, Maven, Gradle, JUnit, Mockito, SonarQube, Postman, Jira, Agile/Scrum, Test Automation, Secure Coding (OWASP), Design Patterns, System Design, High-Availability Engineering

Professional Experience

Software Engineer, *BNY Mellon*

Aug 2024 – Present | Remote, USA

- Built a modular Spring Boot transaction-rules engine enabling dynamic validation flows for multi-currency payments while integrating Dockerized microservices supporting secure governance across treasury operations.
- Implemented Java-based settlement orchestration coordinating batch reconciliation with downstream risk systems, strengthening intraday liquidity visibility and improving audit alignment across corporate banking remittance workflows.
- Developed React interfaces optimizing portfolio-account dashboards with real-time holdings refresh, enhancing trader usability and ensuring consistent UX across complex wealth-management reporting activities.
- Containerized high-throughput payment-processing modules using Docker, improving deployment predictability across shared compute clusters and standardizing runtime configurations for diverse internal clearing applications.
- Optimized Spring Boot API throughput by 36% through connection-pool tuning, caching enhancements, and efficient query patterns supporting time-sensitive financial-instrument pricing services.
- Strengthened exception-handling pipelines with structured logging and event tracing, reducing reconciliation-issue triage time by 27% across credit-operations monitoring environments.

Software Engineer, *Cardinal Health*

Jun 2021 – Jul 2023 | Remote, India

- Built a Spring Boot clinical-order workflow engine coordinating prescription validation, dispensing events, and patient-safety checkpoints across integrated pharmacy-management environments supporting nationwide operations.
- Implemented Java services processing HL7 medication-routes data, strengthening interoperability with EHR systems and improving consistency of administered-therapy updates within patient-centric clinical applications.
- Deployed Kubernetes-orchestrated medication-inventory microservices enabling automated scaling for high-volume hospital requests, improving resilience during seasonal surges across critical pharmaceutical networks.
- Designed Angular dashboards offering unified clinical-supply visibility with responsive filtering, enhancing operational transparency and reducing manual lookup delays across hospital pharmacy coordination processes.
- Improved API response latency by 41% through optimized caching, asynchronous flows, and refined query strategies powering time-sensitive healthcare-fulfillment decisions across distribution centers.
- Enhanced audit-trail completeness by introducing structured compliance logging aligned with HIPAA controls, strengthening security posture for medication-lifecycle activities across regulated healthcare workflows.
- Automated batch-consolidation routines reducing nightly reconciliation duration by 28% through optimized Java scheduling, efficient indexing, and improved microservice coordination across clinical-supply data domains.

Projects

ERP Finance Data Processing Platform

- Developed an end-to-end **financial data management system** integrating **Spring Boot APIs with Oracle DB** to handle invoicing, payment settlements, and transaction logging.
- Reduced query execution time by 45% through optimized indexing and stored procedures.

Financial Insights and Reporting Dashboard

- Designed **Node.js microservices** to aggregate real-time ledger and revenue data; built **REST APIs** for seamless data visualization.
- Improved data refresh time from 2.5s to 0.9s, supporting executive finance analytics.

Education

Master of Science, University at Albany (State University of New York)

Aug 2023 – May 2025

Computer Science